

Ritvik Sharma

New York, NY

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Education

Columbia University

Aug 2025 - Dec 2026

MS in Computer Science - AIML & NLP Track

SRM University – Rank: 2/700+

Jun 2019 - May 2023

BTech in Computer Science - AI & ML Track

GPA: 9.86/10

Work Experience

Columbia University

Oct 2025 - Present

Graduate Research Assistant

New York, USA

- **PeopleScope:** Built a real-time social awareness system using a Meta headset and computer vision (face recognition, emotion, activity and appearance detection) to help BLV users navigate social environments; designed a distributed GPU inference pipeline processing headset camera streams across **3 servers with <100 ms latency**.
- **VisibleLandmarks:** Developed landmark-based navigation using Street View imagery and depth estimation to detect permanent objects (fire hydrants, crosswalks) and estimate precise geocoordinates, enabling BLV users to orient using tactile landmarks and **memorize routes for independent navigation**.

Barclays, UK

Jul 2023 - Jul 2025

Software Engineer

Pune, India

- **Saved \$200K/year** by engineering AWS Lambda optimization utilities and **driving adoption** across Dev/DevOps.
- Achieved **96% faster performance (12 hrs to 27 mins)** for **5M+ record** ETL pipelines by redesigning Glue to DynamoDB workflows using Spark, Dynamic Frames, Batch Write; **Coordinated** with platform teams for seamless deployment.
- **Reduced error-resolution times by 75%** by designing and developing a complex error-routing utility via Python-based Lambdas, IBM Netcool, and SNS filters. Played a **key role in facilitating communication** between development and operations teams to align on system-wide logging and monitoring strategies.
- **Spearheaded the engineering and development** of the integration layer connecting Onboarding, Rewards, Payments, and other modules for Barclays' first cloud-based credit card platform, **ArrowCard** targeting **30M+ UK customers**.

Projects

VELVÉT - Luxury E-Commerce Website — AWS (SageMaker, OpenSearch), React, JavaScript, Python, NLP

[Link to Live Demo](#)

- Built a full-stack luxury e-commerce platform featuring **ML-powered semantic product search** using AWS SageMaker NER models and OpenSearch, enabling natural language product discovery across catalog data.
- Developed scalable backend APIs and React frontend supporting cart operations, recommendations, and product similarity pipelines.

Columbia x Amazon Hackathon Winner: MyCFO.ai — AWS (Bedrock, Api-Gateway), TypeScript

[Link to Web/Video Demo](#)

- Built a **multi-agent financial analysis system** on Amazon Bedrock where a supervisor agent orchestrates specialized agents for forecasting, expense analysis, and investment insights; developed a TypeScript frontend that converts agent outputs into real-time dashboards.

Technical Skills

Languages: Python, Java, JavaScript, C/C++, SQL

Frameworks: React, TypeScript, Node.js, FastAPI, Flask, Django

Cloud & DevOps: AWS (Lambda, SageMaker, OpenSearch), Google GCP, Docker, Kubernetes, Jenkins, Linux

AI / ML: PyTorch, NLP, Computer Vision, Scikit-Learn, Pandas, NumPy, OpenCV

Achievements & Publications

- **2 Research Publications:** (1) **ACM CHI** paper on **PeopleScope**, a real-time social awareness system for BLV users using computer vision and AR headsets; (2) **Springer Nature** NLP paper proposing a retrieval optimization achieving **50x faster inference**.
- **2x Hackathon Winner** in university-level and industry-sponsored competitions, recognized for building innovative, production-grade AI and full-stack solutions under tight deadlines.
- Provide one-on-one tutoring to Columbia student-athletes in Probability, Machine Learning, Calculus, and Data Structures & Algorithms.